



Lab 1: Threat Mapping and Breach Analysis

Mapped COs: CO1, CO4, CO5	Bloom: L4 - Analyze	Total: 100 marks
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Problem Statement	Analyze a real or publicly documented cybersecurity breach and map the attack to security goals, threat categories and cyber kill chain stages. Classify the threat, identify affected assets, estimate risk and prepare a concise, mitigation-oriented analysis report.
Lab Instructions	Use MITRE ATT&CK, NIST CSF and public breach reports as guided by faculty. Show threat classification, kill chain mapping, a risk matrix and mitigation strategies in a structured report.
Tools	MITRE ATT&CK Navigator, NIST CSF reference, public breach reports and CISA advisories, and a documentation/spreadsheet tool.

Running example used throughout this lab: a fictional but realistic ransomware-and-data-theft breach at a mid-sized company, “Acme HealthCare”, whose patient database was accessed after an employee opened a phishing email. Each question applies its algorithm to this same case so you can see the full analysis come together.



Q1

Classify the main threats using the CIA triad and security goals

5 Marks

Objective: Identify which of Confidentiality, Integrity and Availability were violated in the breach and state the security goal affected by each attacker action.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	Read the breach summary and list every attacker action you can identify.	Breach report	List of actions
2	ANALYSE	For each action, decide whether it broke Confidentiality, Integrity or Availability.	CIA triad	C / I / A tag per action
3	ANALYSE	State the specific security goal that failed (e.g. "data secrecy", "data accuracy", "service uptime").	Security goals	Goal named per action
4	REPORT	Summarise which CIA element was most severely impacted overall.	—	One-line verdict
5	REPORT	Record the classification in a small table for the report.	Table template	Completed CIA table

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Applying the steps to Acme HealthCare

The attacker (1) stole 50,000 patient records, (2) encrypted the database with ransomware, and (3) took the booking portal offline. Each action is tagged against the CIA triad:

Attacker action	CIA element broken	Security goal failed
Stole patient records	Confidentiality	Data secrecy
Encrypted the database	Integrity	Data accuracy / access
Booking portal taken offline	Availability	Service uptime

Overall verdict: Confidentiality was most severely impacted, because sensitive medical data left the organisation permanently.



Reusable Template

CIA classification table (copy and fill for your own breach)

Attacker action	CIA element (C / I / A)	Security goal failed
Compromised and locked administrative credentials	Confidentiality/integrity	System control authentication integrity
Encrypted data centre server and storage systems	Integrity	Data accuracy/data trustworthiness
Disabled state wide VPN and took CAT C5 portal office.	Availability	Service uptime and operational continuity

Your Output / Screenshot (paste result here):

West Bengal Police Cyber Crime Wing (CCW) Data Centre Incident (2025)

- i) Attacker compromised administrative access
- ii) Locked out all police officials and staff
- iii) Encrypted the database and central storage system using ransomware
- iv) Disabled the statewide police VPN
- v) Took the CAT C5 tracking portal offline

Overall Verdict: Availability was most severely impacted, because police lost access to tracking software and district networks across the state, halting active investigation..